

AI-Driven Vehicle Accident Analysis System for Automated Damage Identification, Repair Cost Prediction, and Blockchain-Secured Insurance Claim Verification

Research Overview Report

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Abstract

Road accidents continue to cause significant injuries, financial losses, and delays in emergency response and insurance processing worldwide. Existing solutions address individual aspects of accident management in isolation, leaving a critical gap in end-to-end automation. This research presents an AI-driven vehicle accident analysis system that integrates four tightly coupled components: a dedicated IoT-based crash detection device, a multimodal 3D vehicle damage identification module, a region-aware explainable repair cost prediction engine, and a blockchain-secured accident evidence and insurance claim verification layer. Together, these components form a continuous pipeline from physical impact detection through to a tamper-proof, smart-contract-verified insurance claim — without requiring manual intervention at any stage.

1. Introduction

The aftermath of a road accident places simultaneous and conflicting demands on all involved parties. Emergency services must be contacted quickly to prevent loss of life. Insurance companies require accurate, verifiable evidence of damage before processing claims. Repair workshops need objective cost benchmarks to produce fair estimates. And policyholders need assurance that the system is transparent and resistant to manipulation. No existing system satisfies all of these requirements simultaneously.

Current smartphone-based crash detection tools are unreliable in the absence of a charged, properly mounted device and are susceptible to false positives from road vibration or sudden braking. Damage assessment tools predominantly rely on 2D photographic analysis, which cannot capture volumetric deformation or structural damage depth. Repair cost estimation models in commercial use lack transparency and do not account for regional labour and parts pricing variability. Insurance claim systems remain largely manual, document-based, and vulnerable to fraud through document forgery and inflated repair quotes.

This research proposes a unified AI-driven system that addresses all of these shortcomings through four integrated components operating as a single automated pipeline. The system is designed for

deployment in Sri Lanka, where road accident rates are high and insurance fraud is a documented problem, but its architecture is generalisable to other markets.

2. Research Problem

The central research problem is the lack of an integrated intelligent system capable of autonomously managing the complete vehicle accident lifecycle — from crash detection and damage identification through to repair cost prediction and fraud-resistant insurance claim verification — in a single, cohesive, and interoperable framework.

The following specific gaps motivate this research:

- No dedicated, always-on hardware solution exists for accurate crash detection and severity classification that is independent of smartphone availability or driver action
- Existing damage assessment tools are limited to 2D image analysis and cannot measure dent volume, structural deformation depth, or three-dimensional damage extent
- Repair cost estimation models lack regional pricing awareness and provide no explainability, making their outputs difficult to audit or dispute
- There is no tamper-proof, verifiable record-keeping mechanism that connects accident evidence to insurance claim processing in an auditable and fraud-resistant way
- No existing solution integrates all four stages into a pipeline where the verified output of each stage automatically becomes the input for the next

3. Proposed System

The proposed system is structured as a four-stage pipeline. Each stage produces a structured output that is consumed by the following stage, and all outputs are ultimately anchored to an immutable blockchain record. The pipeline is triggered automatically when a crash is detected and proceeds without requiring manual handoffs between stages.

Stage	Input	Output
IoT Crash Detection	Physical crash event	Severity score + GPS coordinates + timestamp
3D Damage Identification	Vehicle images + LiDAR depth maps	3D damage map + affected parts + dent volume
Repair Cost Prediction	Damage data + vehicle + region	Itemised, explainable repair cost estimate
Blockchain Verification	All upstream outputs	Immutable, smart-contract-verified insurance claim

Table 1. Pipeline stages, inputs, and outputs.

4. Component Descriptions

4.1 IoT-Based Crash Detection and Emergency Notification

The first component is a purpose-built IoT device that is permanently installed in the vehicle. It houses an accelerometer, a gyroscope, and a GPS module, and executes embedded firmware that processes sensor data continuously in real time, independently of any smartphone or external application.

On detecting a sudden impact, the firmware applies multi-sensor fusion algorithms to classify the event as minor, moderate, or severe. This severity classification drives a differentiated notification strategy. A minor impact generates a notification to the insurance agent. A moderate or severe impact simultaneously alerts the police, the nearest hospital, and the insurance agent. Each notification payload includes the GPS coordinates of the crash, the computed severity score, and a precise Unix timestamp.

False-positive suppression is implemented at the firmware level. The system cross-validates readings across all onboard sensors to discriminate genuine crash signatures from road bumps, hard braking events, or rough terrain. All confirmed event data is transmitted to a cloud backend via cellular or Wi-Fi for consumption by the damage identification component.

4.2 Multimodal 3D Vehicle Damage Identification

Following crash confirmation, the driver or a first responder uses a mobile application to capture aligned 2D photographs and LiDAR depth maps of the damaged vehicle. This component processes the combined data through a hybrid deep learning architecture that fuses image-based and three-dimensional point cloud analysis.

A convolutional neural network (CNN) trained on labelled vehicle damage datasets analyses the 2D photographs to identify damage categories — dents, cracks, panel deformation, and structural failure — and localises each damage region to a specific vehicle panel. In parallel, a PointNet++ model processes the LiDAR-derived 3D point cloud, comparing it against a reference undamaged model to quantify dent volume and deformation depth in cubic centimetres. The two processing streams are merged through an attention-based fusion module that assigns adaptive weights to each modality based on scene confidence.

The output is a 3D colour-coded damage map annotated with affected panel labels, damage category classifications, and a volumetric severity score. This structured output provides a significantly more objective and measurable basis for cost estimation than qualitative photographic assessments. Training data scarcity is mitigated through generative image augmentation using ControlNet to synthesise realistic damage variations.

4.3 Region-Aware Explainable Repair Cost Prediction

The structured damage output from the identification component is passed to the repair cost prediction module. This component estimates itemised repair costs using an ensemble of machine learning regression models augmented with rule-based pricing logic, and is explicitly designed to account for geographic variation in workshop labour rates and spare parts pricing.

Input features encompass the damage category, affected panel, volumetric severity score, vehicle make, model and year of manufacture, and the geographic region of the nearest authorised repair centre. The model returns a line-item cost breakdown separating parts, labour, surface finishing, and quality assurance costs, together with an explainability report that quantifies each input feature's contribution to the total estimate using established feature importance techniques.

This transparency distinguishes the module from existing black-box prediction tools. Insurance adjusters can audit any estimate and trace every cost line back to a specific damage measurement, reducing disputes and accelerating claim approval cycles.

4.4 Blockchain-Secured Accident Evidence, Legal Compliance, and User Data Protection

The fourth component serves as the trust, legal, and security backbone of the entire system. All outputs from the three upstream components — crash metadata, damage reports, and repair cost estimates — are cryptographically hashed using SHA-256 and stored on a distributed blockchain ledger as immutable records. When a claim is initiated, a smart contract automatically verifies policy coverage, accident timestamps, and cost bounds, then triggers the claim workflow for final insurer review.

Because every record is captured automatically at the time of the accident and cannot be subsequently altered, the blockchain entry constitutes a legally significant evidentiary artefact. This directly supports dispute resolution in court, police investigations, and insurance tribunal proceedings under Sri Lanka's Motor Traffic Act, replacing the current reliance on manually assembled, unverifiable post-accident documentation.

User data security is enforced through an off-chain storage model where only hashes reside on the ledger and all raw data — images, personal identifiers, and policy documents — is held in an AES-256 encrypted database. Role-based access control restricts each stakeholder class (policyholder, insurer, police, and repair centre) to only the data relevant to their function, with all access events logged immutably. This design aligns with data minimisation and purpose-limitation principles under Sri Lanka's Personal Data Protection Act.

5. System Integration

All four components communicate through a cloud-hosted backend using RESTful APIs. The IoT device posts a crash event record immediately upon detection. The damage identification module retrieves this record, appends its 3D assessment output, and posts the enriched record. The cost prediction module reads the damage output, appends its itemised estimate, and commits the full record. The blockchain component subscribes to completed records via an event listener and writes

the final hashed entry to the ledger automatically.

This event-driven, append-only design ensures that each stage operates on complete, verified data from all upstream stages. It also means the blockchain record at the end of the pipeline represents the full, unbroken accident lifecycle — from the first sensor reading at the crash site to the verified repair cost figure — rather than a retrospective reconstruction that could be disputed.

6. Novelty and Contribution

The primary research contribution is the integration of four individually novel components into a single automated, end-to-end pipeline. The system-level integration is itself a research contribution, as no published work addresses the full accident lifecycle — from hardware-level crash detection through to blockchain-verified insurance claim settlement — within a unified framework.

Component-level contributions include the following:

- A dedicated vehicle-mounted IoT crash detection device with embedded multi-sensor fusion and severity-differentiated, multi-stakeholder emergency notification — independent of smartphone availability
- An attention-based fusion architecture combining CNN image feature extraction with PointNet++ 3D point cloud analysis for volumetric vehicle damage measurement
- A region-aware, explainable repair cost prediction engine that links measured dent volume directly to itemised pricing, removing the opacity of existing estimation tools
- A blockchain trust layer providing cryptographically verifiable, tamper-proof accident evidence storage; smart-contract-automated insurance claim verification; legally admissible evidentiary records for dispute resolution and regulatory compliance; and role-based, privacy-preserving access control that protects sensitive user data while maintaining full auditability for all authorised stakeholders

7. Key Technologies

Domain	Technologies
IoT Hardware	Accelerometer, gyroscope, GPS module, cellular/Wi-Fi transceiver, embedded firmware
Computer Vision	YOLOv8, CNN, PointNet++, attention-based LiDAR-image fusion, ControlNet augmentation
Machine Learning	Regression models, rule-based pricing logic, explainable AI (feature importance methods)
Blockchain	Smart contracts, SHA-256 cryptographic hashing, distributed ledger, role-based access control

Backend / Cloud	REST APIs, cloud data storage, event-driven microservices, real-time notification services
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Table 2. Key technologies by domain.

8. Research Context

This research is conducted under the IT4010 Research Project module at the Sri Lanka Institute of Information Technology (SLIIT), within the Software Systems and Technologies (SST) research group, Software Engineering specialisation. The work is motivated by road accident management challenges specific to Sri Lanka, including high accident frequency, delayed emergency response, documented insurance fraud, and the absence of standardised digital workflows connecting accident scenes to insurer systems.

All four components align with the Software Engineering specialisation through the application of modular software architecture, REST API design principles, cloud-based service integration, machine learning model development, embedded firmware engineering, and distributed ledger implementation. The research dataset requirements span IoT sensor readings, annotated vehicle damage images, LiDAR point clouds, regional repair pricing data, and insurance claim records.